

1. **Develop a Swing application that demonstrates the use of the following layout managers: FlowLayout, BorderLayout, and GridLayout. Switch among layouts using radio buttons.**

Code:

```
import javax.swing.*;
import java.awt.*;

public class Q4_LayoutManagerDemo extends JFrame {
    JPanel panel;

    Q4_LayoutManagerDemo() {
        setTitle("Layout Manager Demo");
        setSize(600, 400);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        JPanel top = new JPanel();
        JRadioButton flow = new JRadioButton("FlowLayout", true);
        JRadioButton border = new JRadioButton("BorderLayout");
        JRadioButton grid = new JRadioButton("GridLayout");

        ButtonGroup group = new ButtonGroup();
        group.add(flow);
        group.add(border);
        group.add(grid);

        top.add(flow);
        top.add(border);
        top.add(grid);
        add(top, BorderLayout.NORTH);

        panel = new JPanel();
        add(panel, BorderLayout.CENTER);

        flow.addActionListener(e -> changeLayout("flow"));
        border.addActionListener(e -> changeLayout("border"));
        grid.addActionListener(e -> changeLayout("grid"));

        changeLayout("flow");
    }

    void changeLayout(String type) {
        panel.removeAll();

        if (type.equals("flow")) {
            panel.setLayout(new FlowLayout());
            for (int i = 1; i <= 5; i++) panel.add(new JButton("Button " + i));
        } else if (type.equals("border")) {
            panel.setLayout(new BorderLayout());
            panel.add(new JButton("North"), BorderLayout.NORTH);
            panel.add(new JButton("South"), BorderLayout.SOUTH);
        }
    }
}
```

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        panel.add(new JButton("East"), BorderLayout.EAST);
        panel.add(new JButton("West"), BorderLayout.WEST);
        panel.add(new JButton("Center"), BorderLayout.CENTER);
    } else {
        panel.setLayout(new GridLayout(2, 3, 10, 10));
        for (int i = 1; i <= 6; i++) panel.add(new JButton("Button " + i));
    }

    panel.revalidate();
    panel.repaint();
}

public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> new Q4_LayoutManagerDemo().setVisible(true));
}
}

```

Output:

